

**1. AIM**

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To design an CMOS inverter repeater using minimum channel length and minimum width in 180nm technology and determine

- 1]  $R_{out}$  , (output resistance of the inverter)
- 2]  $C_{out}$  , (output capacitance of the inverter)
- 3] intrinsic delay  $t_p$  , (intrinsic delay of the inverter)
- 4]  $R_n$  ,  $R_p$  (resistance of NMOS and PMOS)
- 5]  $k$  (number of stage of repeaters)

to calculate propagation delay without inverter/buffer insertion and with inverter/buffer insertion and to compare the results.

**2. TOOL**

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Cadence Virtuoso

**3. THEORY**

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1. Unbuffered Interconnect Delay: A long interconnect exhibits distributed resistance  $R_{int}$  and capacitance  $C_{int}$ , causing propagation delay to scale quadratically with wire length. Driven by a minimum-sized inverter ( $R_{out}$ ,  $C_{out}$ ), the total unbuffered delay is given by  $T_{50\%} = 0.7R_{out}(C_{int} + C_{out}) + R_{int}(0.4C_{int} + 0.7C_{out})$ .

2. Minimum-Sized Repeater Insertion: Inserting intermediate minimum-sized inverters divides the line into  $k$  shorter sections, converting quadratic delay growth into a linear relationship. The total propagation delay for  $k$  minimum-sized repeater stages is given by  $T_{50\%} = k \left[ 0.7R_{out} \left( \frac{C_{int}}{k} + C_{out} \right) + \frac{R_{int}}{k} \left( 0.4 \frac{C_{int}}{k} + 0.7C_{out} \right) \right]$ .

3. Optimum Number of Stages: Minimizing the total propagation delay with respect to  $k$  yields the optimum number of repeater stages:  $k = \sqrt{\frac{0.4R_{int}C_{int}}{0.7R_{out}C_{out}}}$ .

## 4. TABLES

**Table 4.1: Design Parameters**

| given parameter       | value | unit                     |
|-----------------------|-------|--------------------------|
| technology            | 180   | nm                       |
| $V_{dd}$              | 1.8   | V                        |
| $V_{tn}$              | 0.45  | V                        |
| $ V_{tp} $            | 0.45  | V                        |
| minimum width of nmos | 400   | nm                       |
| minimum width of pmos | 400   | nm                       |
| $k_n'$                | 70    | $\mu\text{A}/\text{V}^2$ |
| $k_p'$                | 20    | $\mu\text{A}/\text{V}^2$ |
| $R_{int}$             | 68    | $\text{k}\Omega$         |
| $C_{int}$             | 60    | fF                       |

**Table 4.2: Derived and observed intrinsic parameters**

| derived and observed parameter  | value    | unit             |
|---------------------------------|----------|------------------|
| $t_{phl}$                       | 9.8587   | ps               |
| $t_{plh}$                       | 27.33919 | ps               |
| $t_p$                           | 18.5989  | ps               |
| $R_n$                           | 4.762    | $\text{k}\Omega$ |
| $R_p$                           | 16.67    | $\text{k}\Omega$ |
| $R_{out}$                       | 10.716   | $\text{k}\Omega$ |
| $C_n$                           | 3.0004   | fF               |
| $C_p$                           | 2.3768   | fF               |
| $C_{out}$                       | 3.0004   | fF               |
| $k$ (number of repeater stages) | 8        | unitless         |

**Table 4.3: Result**

| parameter | without inv/buf insertion | with inv/buf insertion | unit | error     |
|-----------|---------------------------|------------------------|------|-----------|
| $t_{phl}$ | 2.882                     | 1.17325                | ns   | -59.2904% |
| $t_{plh}$ | 3.119                     | 1.23534                | ns   | -60.3931% |
| $t_p$     | 3.0005                    | 1.2043                 | ns   | -59.8634% |

**Table 4.4: comparison of theoretical and practical values**

| parameter                 | theoretical value ( $T_p$ ) | practical value ( $t_p$ ) | unit | signed error |
|---------------------------|-----------------------------|---------------------------|------|--------------|
| without inv/buf insertion | 2.2473                      | 3.0005                    | ns   | +33.5158%    |
| with inv/buf insertion    | 0.9768                      | 1.2043                    | ns   | +23.2903%    |

## 5. CALCULATION

$$\begin{aligned}
 R_n &= 1000 / (k_n * (W/L) * (V_{dd} - V_{tn})) \\
 &= 1000 / (70 * (400/180) * (1.8 - 0.45)) \\
 &= 4.7619048k\Omega \approx 4.762k\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_p &= 1000 / (k_p * (W/L) * (V_{dd} - V_{tp})) \\
 &= 1000 / (20 * (400/180) * (1.8 - 0.45)) \\
 &= 16.6666667k\Omega \approx 16.67k\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_{out} &= (R_n + R_p) / 2 \\
 &= (4.7619048 + 16.6666667) / 2 \\
 &= 10.7142858k\Omega \approx 10.714k\Omega
 \end{aligned}$$

$$\begin{aligned}
 C_{out1} &= t_{phl} / (0.69 * R_n) \\
 &= 9.8587 / (0.69 * 4.7619048) \\
 &= 3.0004739ff \approx 3ff
 \end{aligned}$$

$$\begin{aligned}
 C_{out2} &= t_{plh} / (0.69 * R_p) \\
 &= 27.33919 / (0.69 * 16.6666667) \\
 &= 2.3773209ff \approx 2.3773ff
 \end{aligned}$$

$$\begin{aligned}
 C_{out} &= \max(C_{out1}, C_{out2}) \\
 &= \max(3.0004739, 2.3773209) \\
 &= 3.0004739ff \approx 3ff
 \end{aligned}$$

$$\begin{aligned}
 t_{p(min)} &= (t_{phl} + t_{plh}) / 2 \\
 &= (9.8587 + 27.33919) / 2
 \end{aligned}$$

$$= 18.598945ps \approx 18.5989ps$$

$$k = \sqrt{(R_{int} * C_{int} * 0.4) / (R_{out} * C_{out} * 0.7)}$$

$$= \sqrt{(68 * 60 * 0.4) / (10.7142858 * 3.0004739 * 0.7)}$$

$$= 8.5159777 \approx 8$$

$$T_{p(no\_rep)(theoretical)} = (0.7 * R_{out} * (C_{int} + C_{out})$$

$$+ R_{int} * (0.4 * C_{int} + 0.7 * C_{out})) / 1000$$

$$= (0.7 * 10.7142858 * (60 + 3.0004739)$$

$$+ 68 * (0.4 * 60 + 0.7 * 3.0004739)) / 1000$$

$$= 2.2473261ns \approx 2.2473ns$$

$$T_{p(rep)(theoretical)} = (k * (0.7 * R_{out} * (C_{int}/k + C_{out})$$

$$+ (R_{int}/k) * (0.4 * C_{int}/k + 0.7 * C_{out}))) / 1000$$

$$= (8 * (0.7 * 10.7142858 * (60/8 + 3.0004739)$$

$$+ (68/8) * (0.4 * 60/8 + 0.7 * 3.0004739))) / 1000$$

$$= 0.976851ns \approx 0.9769ns$$

$$t_{HL(no\_rep)} = 2.882ns$$

$$t_{LH(no\_rep)} = 3.119ns$$

$$t_{p(no\_rep)} = (t_{phl\_no\_rep} + t_{plh\_no\_rep}) / 2$$

$$= (2.882 + 3.119) / 2$$

$$= 3.0005ns$$

$$t_{HL(rep)} = 1.17325ns$$

$$t_{LH(rep)} = 1.23534ns$$

$$t_{p(rep)} = (t_{phl\_rep} + t_{plh\_rep}) / 2$$

$$= (1.17325 + 1.23534) / 2$$

$$= 1.204295ns \approx 1.2043ns$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

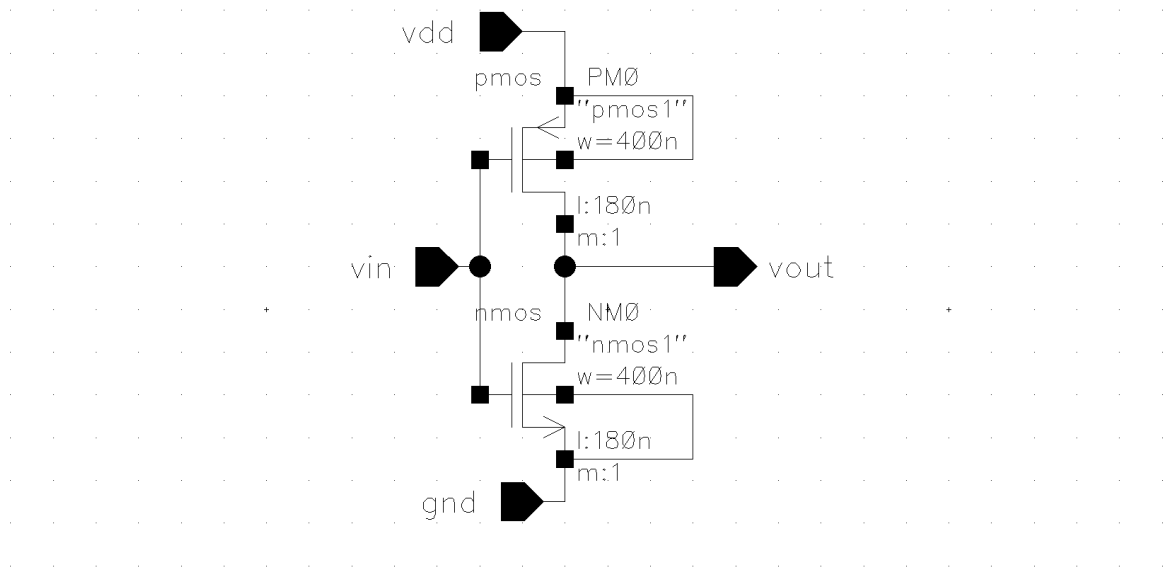


Figure 6.1: Schematic of minimum size inverter

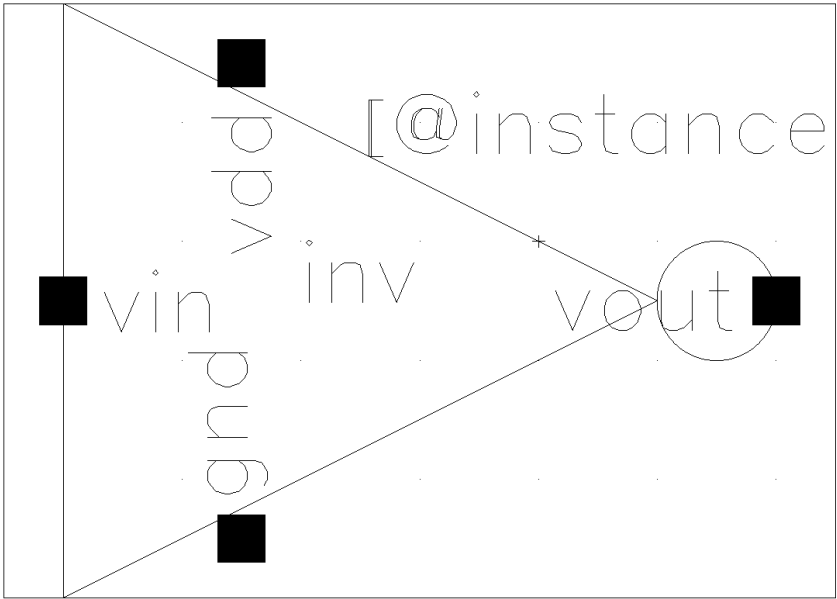


Figure 6.2: Symbol of minimum size inverter

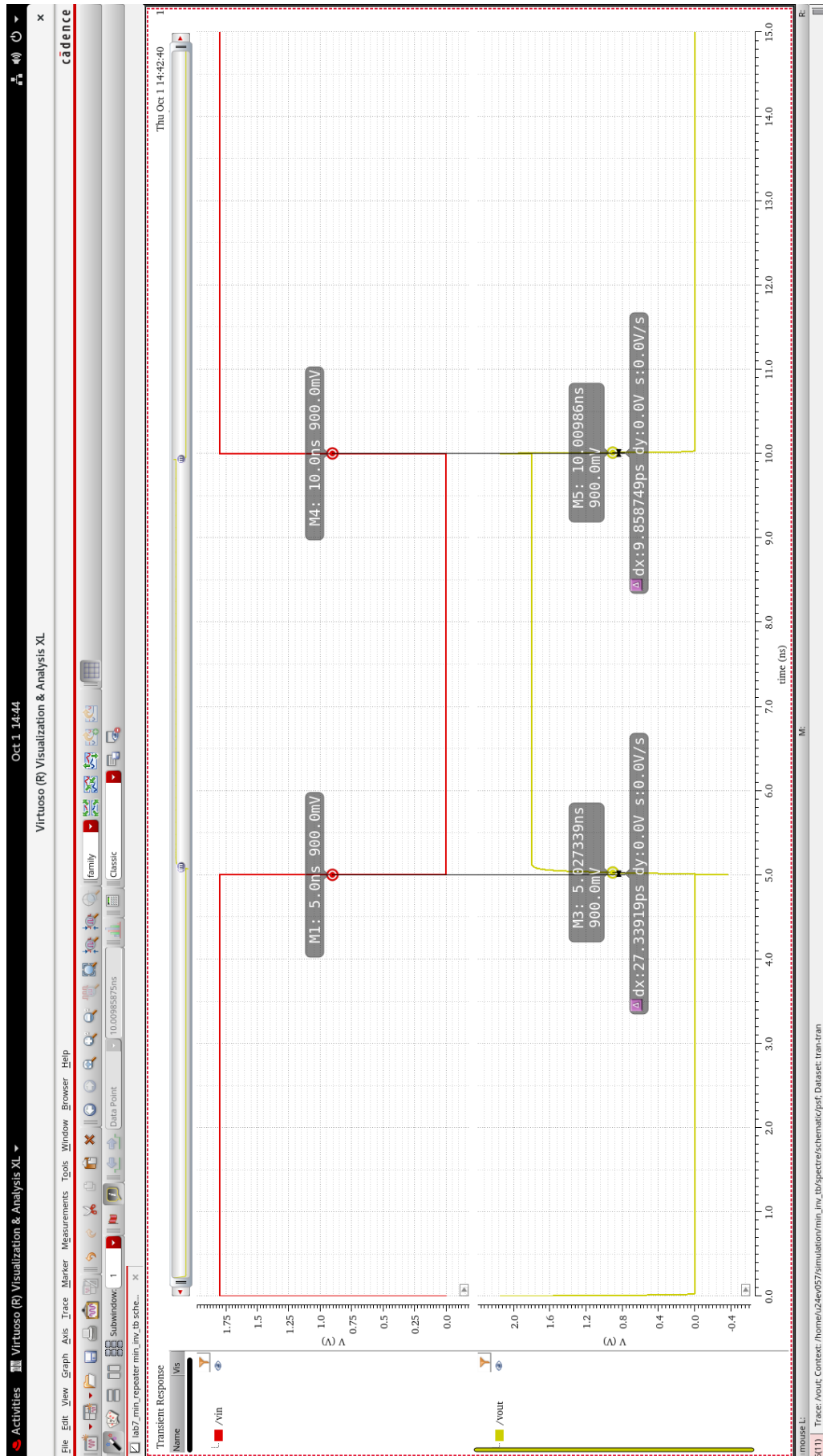


Figure 6.3: Transient waveform of minimum size inverter without load

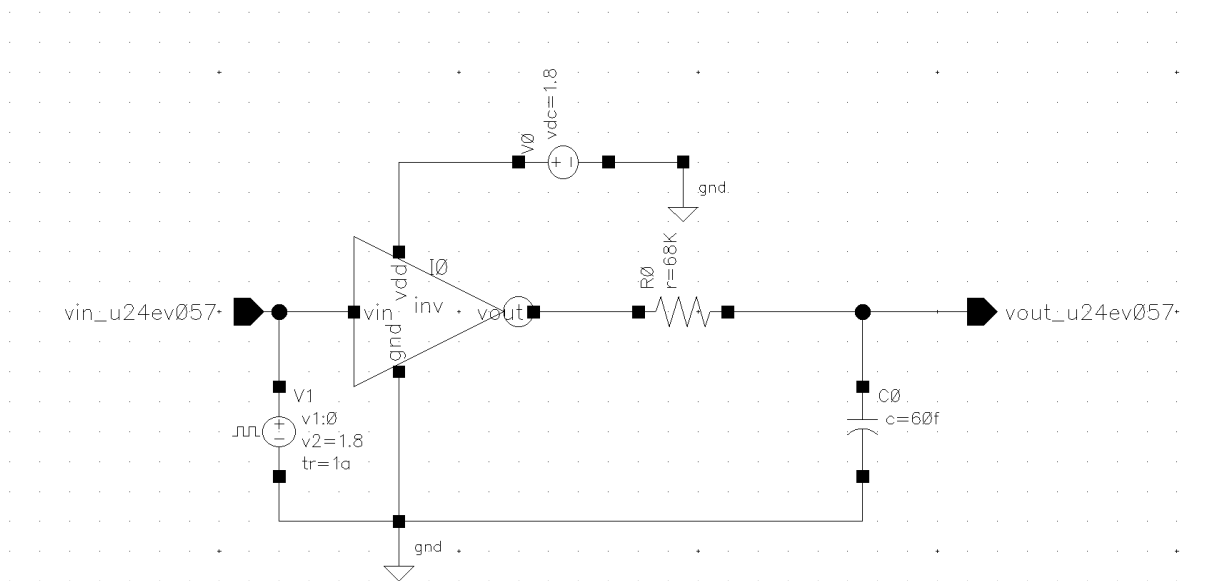


Figure 6.4: Schematic of inverter without inv/buf insertion (without repeaters)

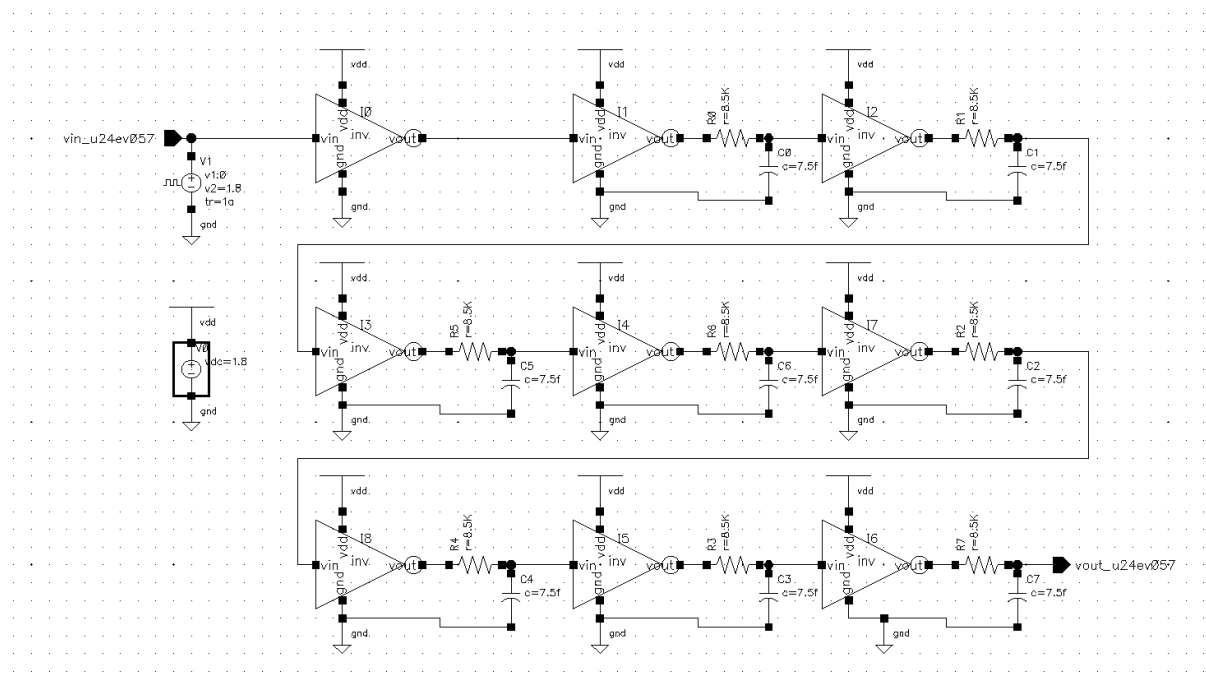
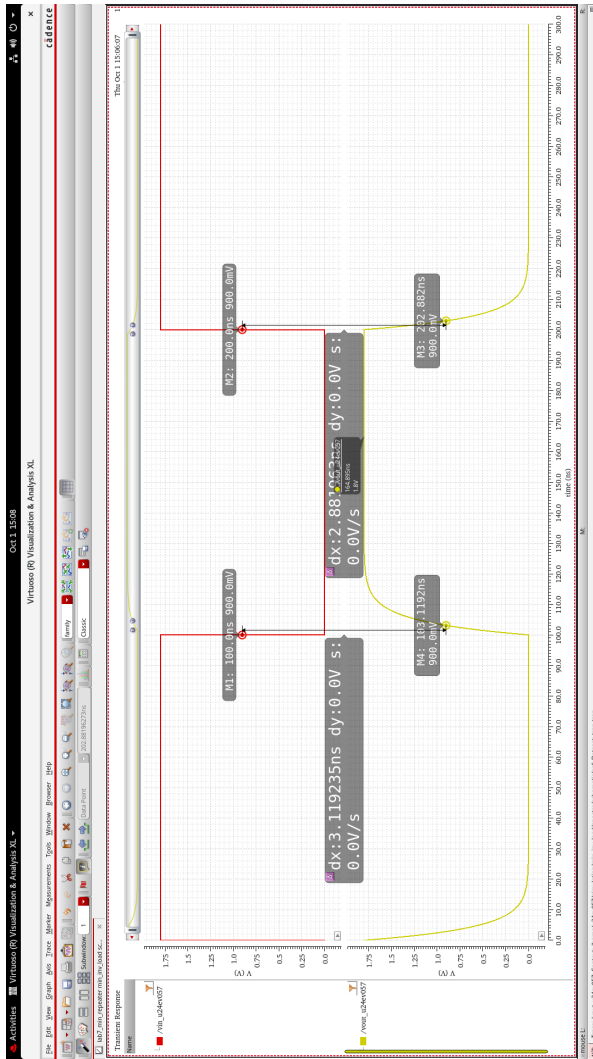
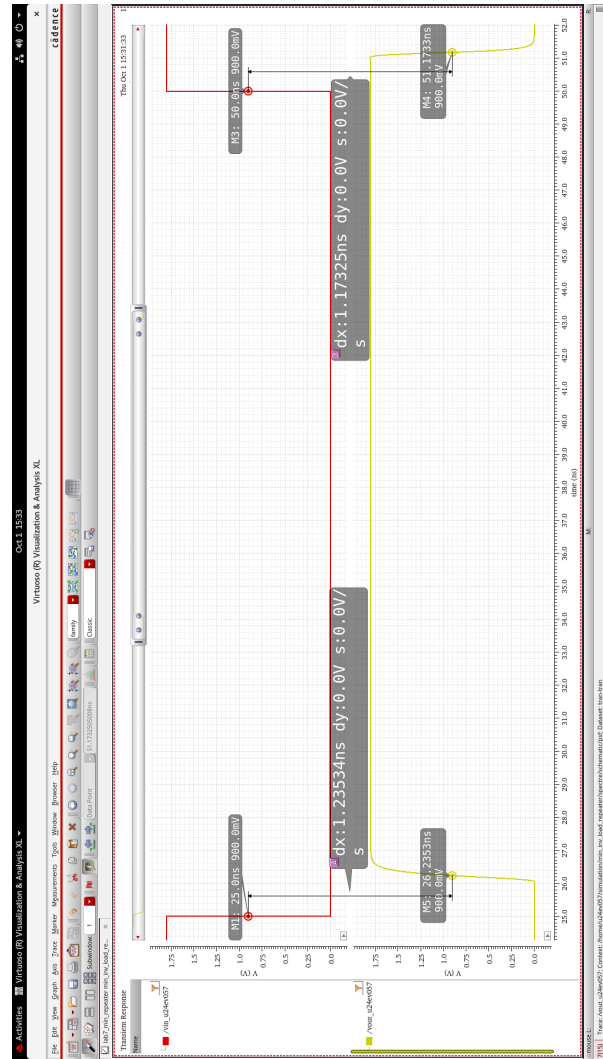


Figure 6.5: Schematic of inverter with inv/buf insertion (with repeaters)



**Figure 6.6: Transient waveform of inverter without inv/buf insertion (without repeaters)**  
 $t_{PHL} = 2.882\text{ns}$ ,  $t_{PLH} = 3.119\text{ns}$ ,  $t_p = 3.0005\text{ns}$



**Figure 6.7: Transient waveform of inverter with inv/buf insertion (with repeaters)**  
 $t_{PHL} = 1.17325\text{ns}$ ,  $t_{PLH} = 1.23534\text{ns}$ ,  $t_p = 1.2043\text{ns}$



## 7. CONCLUSION

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1. An unbuffered interconnect and minimum-sized CMOS inverter repeaters were designed and simulated in 180 nm technology ( $V_{DD} = 1.8\text{V}$ ,  $R_{int} = 68\text{k}\Omega$ ,  $C_{int} = 60\text{fF}$ ).
2. Minimum-sized repeater insertion ( $k = 8$ ) reduced propagation delay from  $3.0005\text{ns}$  down to  $1.2043\text{ns}$  (59.86% reduction), in good agreement with the theoretical delay of  $0.9768\text{ns}$ .